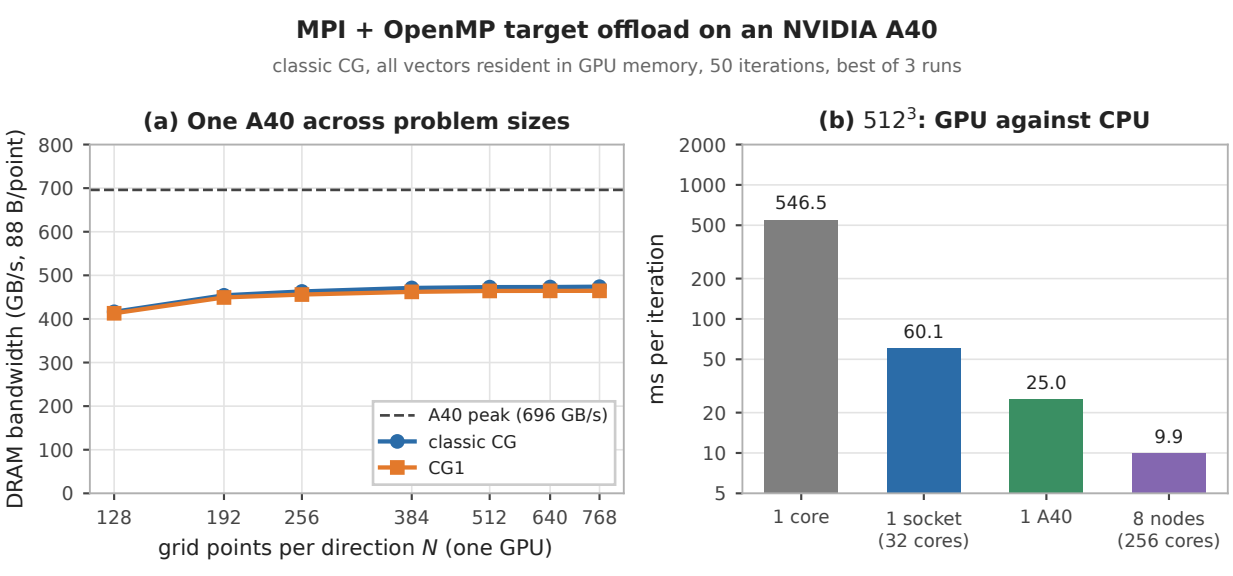
MPI + OpenMP target offload on an NVIDIA A40
classic CG, all vectors resident in GPU memory, 50 iterations, best of 3 runs
(a) One A40 across problem sizes
DRAM bandwidth (GB/s, 88 B/point)
800
700
600
500
400
300
200
100
0
A40 peak (696 GB/s)
classic CG
CG1
128
192
256
384
512
640
768
grid points per direction N (one GPU)
(b) 512³: GPU against CPU
ms per iteration
2000
1000
500
200
100
50
20
10
5
546.5
60.1
25.0
9.9
1 core
1 socket
(32 cores)
1 A40
8 nodes
(256 cores)